

**TABLE II**  
**Active vs Passive Tags**

|              | Advantages                                                                                                                                                                                                                                                                  | Disadvantages                                                                                                                                                                                                                                                                                                                                                               | Remarks                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Passive      | <ul style="list-style-type: none"> <li>• Longer life time</li> <li>• Wider range of form factors</li> <li>• Tags are more mechanically flexible</li> <li>• Lowest cost</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>• Distance limited to 4-5 metres (UHF)</li> <li>• Strictly controlled by local regulations</li> </ul>                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Most widely used in RFID applications</li> <li>• Tags are LF, HF or UHF</li> </ul>                                                  |
| Semi-passive | <ul style="list-style-type: none"> <li>• Greater communication distance</li> <li>• Can be used to manage other devices like temperature sensors, pressure sensors, etc</li> <li>• Do not fall under the same strict power regulations imposed on passive devices</li> </ul> | <ul style="list-style-type: none"> <li>• Expensive due to battery and tag packaging</li> <li>• Reliability—impossible to determine whether a battery is good or bad, particularly in multiple transponder environments</li> <li>• Widespread proliferation of active transponders presents an environmental hazard from potentially toxic chemicals in batteries</li> </ul> | <ul style="list-style-type: none"> <li>• Used mainly in real-time systems to track high-value materials or equipment throughout a factory</li> <li>• Tags are UHF</li> </ul> |
| Active       |                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• Used in logistics for tracking of containers on trains, trucks, etc</li> <li>• Tags are UHF or microwave</li> </ul>                 |

the read-only tag, the microchip or memory is written only once, during manufacturing process. The information, along with the serial number on the read-only tag, can never be changed. In the read-write tag, only the serial number is written during manufacturing process. The remaining blocks can be re-written by the user.

Until recently, the focus of RFID technology was mainly on tags and readers which were being used in systems involving relatively low volumes of data. This is now changing as RFID in the supply chain is expected to generate huge volumes of data, which will have to be filtered and routed to the backend IT systems. To solve this problem, companies have developed special software packages called savants, which act as buffers between the RFID front-end and the IT backend. Savants are equivalent to middleware in the IT industry.

**RFID reader.** RFID reader is the device used to transmit to and receive information from the RFID tag. It is also referred to as an 'interrogator.' It includes sensors that read the RFID tags in the vicinity.

The reader sends a request for information to the tag. The tag responds with the respective information, which the reader then forwards to the data processing device. The tag and reader communicate with one another over a radio frequency channel. In some systems, the link between the reader and the computer is wireless.

**Supporting infrastructure.** The supporting infrastructure includes related software and hardware required for RFID systems. The software manages the interaction between the RFID reader and the RFID tags.

### Communication protocol

The communication process between the reader and tag is managed and controlled by one of several protocols, such as the ISO 15693 and ISO 18000-3 for HF, or the ISO 18000-6 and EPC 18000-6 for UHF. Basically, when the reader is switched on, it starts emitting a signal at the selected frequency band (typically 860-915 MHz for UHF or 13.56 MHz for HF). Any corresponding tag in the vicinity of the reader will detect the signal and use the energy from it to wake up and supply operating power to its internal circuits. Once the tag has encoded the signal as valid, it replies to the reader, and indicates its presence by modulating (affecting) the reader field.

**Anti-collision.** If many tags are present, they will all reply at the same time. At the reader end, this is seen as signal collision and an indication of multiple tags. The reader manages this problem by using an anti-collision algorithm that allows tags to be sorted and individually selected. There are many different types of algorithms (binary tree, aloha, etc) which are defined as part of the protocol standards.

The number of tags that can be identified depends on the frequency

and protocol used, and can typically range from 50 tags/s for HF to 200 tags/s for UHF. Once a tag is selected, the reader is able to perform a number of operations such as reading the tags. This process continues under control of the anti-collision algorithm until all the tags have been selected.

**Inductively coupled RFID tags.** These original tags were complex systems of metal coils, antennae and glass. Inductively coupled RFID tags were powered by a magnetic field generated by the RFID reader. Electrical current has an electrical component and a magnetic component, i.e., it is electromagnetic. The name 'inductively coupled' comes from the magnetic field induced by a current in the wire.

**Capacitively coupled tags.** These tags were created to lower the technology's cost. These were disposable tags that could be applied to less expensive merchandise and made as universal as bar codes. Capacitively coupled tags used conductive carbon ink instead of metal coils to transmit data. The ink was printed on paper labels and scanned by readers.

**Motorola's BiStatix RFID tags.** These were the frontrunners in this technology. They used a silicon chip that was only 3 millimetres wide and stored 96 bits of information. This technology didn't catch on with retailers, and BiStatix was shut down in 2001.

Inductively coupled and capacitively coupled RFID tags aren't used as commonly today because these are ex-